

Analysis and Theosis Report: Relational Continuity as an Executable Control Contract

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1. Abstract

"Theosis" is retained as the requested report title. Here it denotes the architecture's proposed progression toward coherent, self-maintaining operation; it is not treated as a measured scientific property. The technical object of study is whether stable logical identity, explicit authority and durable evidence can coordinate independently replaceable runtime carriers.

2. Research question and hypothesis

Question: can the existing components support bounded self-recovery while preserving identity and durable context?

Hypothesis H1: if ownership is transactionally fenced, snapshots are validated and carrier capabilities are observed, replacing a failed carrier can preserve logical identity without asserting preservation of volatile execution state. H1 is a development hypothesis. The local evidence qualifies constituent operations rather than proving the integrated hypothesis.

3. Architecture decomposition

Identity belongs to a versioned logical record. Authority belongs to an owner/worker/generation/lease binding. Execution belongs to Node, Python, native C, WASM or Chromium. Transport belongs to the selected physical link. Recovery belongs to a verified durable checkpoint plus runtime descriptor. Observation belongs to timestamped evidence. Separating these roles prevents an address declaration, a compiled artifact and an accepted external operation from being conflated.

4. State mechanics

Initial progression is STARTING to RECONNECTING to ACTIVE after worker readiness, client acknowledgment and first-frame verification. Missed heartbeats lead to SUSPECT and frozen input. A valid renewal may restore the previous state. Expiration permits one atomic claimant to increment generation and enter RECOVERING, then RESTORING. A verified restoration enters RECONNECTING. An authenticated fresh transport and client reconciliation permit ACTIVE. Failure is recorded as FAILED; intentional termination as CLOSED. Claims against terminal sessions are rejected.

Lease and generation checks protect database mutations. They do not reverse a web transaction already executed by an old browser. A receiver must validate generation and ownership at dispatch, and the external operation may still require application-specific idempotency or outcome reconciliation. The uncertainty is retained as evidence. Client clocks do not confer lease authority; the database supplies authoritative time.

5. Continuity boundaries and counterexamples

A stable KEX address persists while a UDP socket or WebRTC connection is replaced. A profile checkpoint can restore eligible browser storage and saved tabs; it does not reconstruct arbitrary JavaScript heap state. A shared-memory mapping connects processes on its supported host; it does not become a cross-host transport by naming it logically. A successful local norm calculation does not validate a multi-host mining cluster. These boundaries give concrete counterexamples against an overly broad continuity claim and define useful tests for H1.

The proposed complete continuity contract is identity continuity, durable-data continuity and explicit interruption/reconciliation of execution and transport. Recovery objectives must be measured from failure detection to first useful restored operation, with checkpoint age recorded as potential data loss exposure.

6. Evidence interpretation

E01 exercises idempotence, explicit supersession, conflict retention, zero-address rejection and supervisor receipts. These directly support governance primitives. E02 confirms declared MCP tools and local manifest checks. E03 establishes the arithmetic consistency of one memory map, not synchronization under load. E05 executes WASM exports and selected mathematical/ABI tests; E06 independently exercises three norm examples. E04 initially supplied a falsifier of native build readiness; the corrected build now passes. Runtime operation remains a separate boundary.

The initial E05 inventory labeled an unlocated Rust artifact as `VERIFIED_ESTATE_EXECUTION` while reporting size zero and `UNLOCATED_ON_DISK`. The custody routine has been corrected to emit `UNVERIFIED_MISSING_ARTIFACT` for absent files. The seven case passes must therefore not be generalized to verification of that Rust artifact. The local labeling defect is resolved; execution-qualified attribution for present artifacts remains a broader T09/T10 requirement. The corrected current generated receipt is retained alongside the execution log.

7. Research design for integrated qualification

Experiment A: contend two independent PostgreSQL clients for one expired session; observe winner, generation and stale-write rejection. Experiment B: crash a worker during an unacknowledged input; verify no automatic replay and document the unresolved external outcome. Experiment C: corrupt a checkpoint and prove recovery refuses activation. Experiment D: remove a carrier capability and verify logical identity survives with an explicit readiness transition. Experiment E: deploy an incompatible ABI to a single sector and verify rejection/rollback. Experiment F: lose telemetry and prove the system reports unknown rather than invented measurements.

Record distributions rather than only means: recovery duration, checkpoint age, frame latency, queue occupancy and repair attempts. Compare fault-free and fault-injected workloads with the same release and configuration. Do not infer one-way latency from unsynchronized clocks.

8. Sources and theoretical disposition

Source references: paths above are relative to the stated GitHub baseline. Evidence E01-E06 is supplied in evidence/ with commands, exit codes and SHA-256 receipts. E05-generated-receipt.json is the current generated receipt; original repository evidence was restored after the test. The separate session-control module is outside the source baseline and is not represented as deployed. Source inspection establishes structure, not operating performance. No third-party literature review, standards certification, live pool share acceptance or deployed recovery experiment is claimed.

The research supports a falsifiable programme for self-supporting architecture. It does not establish an emergent intelligence, unbounded autonomy, certification or universal continuity. Its value is a precise bridge from the project's relational vocabulary to observable execution contracts.